

Revised Generation Interconnection System Impact Study Report

For Queue Project Q14-0406

Kensington 138 kV
August 22, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0406. The Interconnection Customer proposes a 223.6 MW solar photovoltaic facility interconnecting at the Kensington 138 kV substation in the Maryland area. The request was received on December 01, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 195 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,373,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Bells Mill to Beverly Farms 138 KV	Tiger Bay 345/138 kV Transformer Bank	159.9	212	10.3
Kensington to Linden 138 KV	Kensington to Linden 138 KV #1	147.4	315	24.5
Kensington to Linden 138 KV	Kensington to Linden 138 KV	138.8	335	24.5
Beverly Farms to Phillips Pump 1 138 KV line	Air Force Base 345 kV/138 kV Auto	120.3	250	7.4
Jerome to Fairland 138 KV	Siuc to Fairland 138 KV	109.2	277	7.4
Darien to Sligo 138 KV L	Taber to Glenannan 345 KV	106.5	366	15.6
Daien to Colsville 138 KV line	Taber to Glenannan 345 KV	105.7	487	13.6
Linden to Sligo 138 KV	Gleannan to Taber 345 KV L	104.9	398	15.6
Bells Mill to Twinbrook 138 KV line	Radfords Run Wind Farm to Haddam Neck 138 KV	101.4	220	7.0
Dixleta to N Pole 138 KV	Taber to Glenannan 345 KV	100.9	472	6.6
North Pole to High Ridge 138 KV	Taber to Dixleta 345 KV	100.3	799	10.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Bells Mill to Beverly Farms 138 KV	Rebuild line with higher-capacity conductor	\$1,535,000	31% (\$481,000)
Kensington to Linden 138 KV Circuit 2	Rebuild line with higher-capacity conductor	\$1,787,000	100% (\$1,787,000)
Kensington to Linden 138 KV Line	Rebuild line with higher-capacity conductor	\$1,514,000	100% (\$1,514,000)
Beverly Farms to Phillips Pump 1 138 KV	Rebuild line with higher-capacity conductor	\$3,555,000	34% (\$1,220,000)
Jerome to Fairland 138 KV	Reconductor line	\$1,518,000	30% (\$456,000)
Darien to Sligo 138 KV L	Reconductor line	\$1,572,000	32% (\$497,000)
Daien to Colsville 138 KV line	Reconductor line	\$363,000	14% (\$51,000)
Linden to Sligo 138 KV	Reconductor line	\$309,000	32% (\$98,000)
Bells Mill to Twinbrook 138 KV line	Reconductor line	\$803,000	30% (\$242,000)
N Pole to Dixleta 138 KV	Reconductor line	\$367,000	2% (\$9,000)
High Ridge to N Pole 138 KV L	Reconductor line	\$865,000	2% (\$18,000)
	Total allocated to project		\$6,373,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.